Dragon Ball × Quillon

FPGA Collaboration Proposal

Response to Chong's Technical Questions

April 10, 2026 — Quillon Foundation

Dragon Ball has proposed collaborating on the FPGA prototype phase of the QUG-V1 mining SoC. This document responds to Chong's specific technical questions about (1) which algorithm the FPGA validates, (2) whether Xilinx Artix-7 has sufficient resources, and (3) how Dragon Ball can contribute to the FPGA phase. We propose a two-stage FPGA validation with Dragon Ball providing device selection expertise and FPGA manufacturing experience, while Quillon provides the complete RTL design and algorithm firmware.

Question 1: Which Algorithm Does the FPGA Validate?

Answer: Both algorithms — but in two stages within Phase 1.

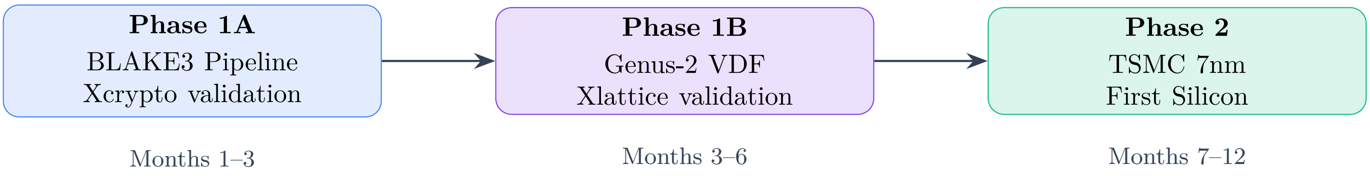
Phase 1A: BLAKE3 Validation (Months 1–3)

- Validate the **Xcrypto BLAKE3 pipeline** (100 sequential hashes per nonce)
- Single-issue pipelined hash unit — the simpler datapath
- **Purpose:** Prove the mining core works, validate hashrate/watt, test 16-core cluster interconnect
- This is the same class of problem as Dragon Ball's ALPH miner — **familiar territory**

Phase 1B: Genus-2 VDF Validation (Months 3–6)

- Validate the **Xlattice bignum accelerator** (256-bit modular arithmetic)
- Requires: Barrett reduction, modular multiplication, modular inversion (Fermat's method)
- **Purpose:** Prove the VDF acceleration works, measure doublings/sec/watt
- This is the **novel part** — no one has done Genus-2 Jacobian hardware acceleration before

Phase 1A can start **immediately**. The Genus-2 VDF algorithm is expected to activate on mainnet in Q3 2026, aligning perfectly with the Phase 1B FPGA validation timeline. Dragon Ball can evaluate the QUG-V1's performance on *both* mining paths during the prototype phase.



Question 2: Artix-7 Resource Assessment

Chong's concern is correct. The Xilinx Artix-7 (XC7A200T) is likely sufficient for Phase 1A but **insufficient for Phase 1B**.

Artix-7 XC7A200T Resources

Resource	Available	Notes
Logic Cells	215,360	Adequate for single-core BLAKE3
DSP48E1 Slices	740	**Bottleneck for bignum**
Block RAM (36Kb)	365	13.14 Mb total
I/O Pins	500	Sufficient

Resource Estimates by Subsystem

Subsystem	LUTs	DSPs	BRAM	Fits Artix-7?
BLAKE3 Xcrypto pipeline (1 core)	~5K	8	4	Yes
RISC-V core (RV32IMC)	~8K	4	8	Yes
Bus + memory controller	~3K	0	16	Yes
Phase 1A total (1 mining core)	**~16K**	**12**	**28**	**Yes (7% util)**
256×256 modular multiplier	~20K	64–128	4	Tight
Barrett reduction unit	~8K	32	2	Tight
8-butterfly parallel Xlattice	~80K	512+	16	**No — exceeds DSPs**
Phase 1B total (full SoC)	**~150K**	**400+**	**64**	**Needs larger FPGA**

The 8-butterfly parallel Xlattice unit requires 512+ DSP slices, which exceeds the Artix-7's 740 DSP budget when combined with the rest of the SoC. Dragon Ball's experience with the Xilinx 355T (likely Kintex-7 or Kintex UltraScale) for their previous FPGA prototypes is directly applicable here. We should select the FPGA device **together** based on Dragon Ball's manufacturing experience.

Recommended FPGA Targets

Validation Target	Recommended FPGA	Purpose	Est. Cost
BLAKE3 core only	Artix-7 200T	Phase 1A — hash pipeline	$500/board
Full SoC (1 core)	Kintex-7 355T or KU040	Phase 1B — complete mining core	$2K–4K/board
Multi-core cluster (4 cores)	Virtex UltraScale+	Optional — interconnect validation	$8K–15K/board

Direct Answer to Chong's Question

Phase 1A (BLAKE3 only): Yes, Artix-7 A200T has sufficient resources. Our synthesis estimates show ∼16K LUTs, 12 DSPs, and 28 BRAM blocks — just 7% utilization. Artix-7 is viable and low cost for this phase.

Phase 1B (Full SoC with Xlattice): No, Artix-7 is insufficient. The 8-butterfly Xlattice unit alone needs 512+ DSPs, exceeding Artix-7's 740 budget when combined with the rest of the SoC. **Chong is correct** that a larger device is needed.

Our Device Recommendation

For Phase 1B, we recommend the **Xilinx Kintex-7 XC7K325T-2FFG900I**:

Resource	Kintex-7 325T	QUG-V1 Need	Margin
LUTs	203,800	∼150K	35% headroom
DSP48E1	840	∼550	35% headroom
Block RAM	16.02 Mb	∼10 Mb	60% headroom
Clock target	200–300 MHz	100 MHz (prototype)	Comfortable

This part provides ample headroom, is in the same Kintex-7 family as Dragon Ball's existing 355T boards (toolchains and board designs may be reusable), and costs $2K–4K per development board. We welcome Dragon Ball's confirmation or alternative suggestion based on your inventory.

Question 3: FPGA Collaboration Proposal

Yes — we warmly welcome Dragon Ball's involvement in the FPGA phase. This strengthens both parties: Dragon Ball gains early insight into the algorithm, and Quillon benefits from Dragon Ball's FPGA manufacturing and verification expertise.

Proposed Division of Work

Quillon Provides	Dragon Ball Provides
Complete RTL design (Verilog/SystemVerilog) for QUG-V1 SoC	FPGA device selection expertise
Mining algorithm firmware (BLAKE3 + Genus-2 VDF)	Board design and/or existing dev board access
Testbench and verification suite	FPGA synthesis optimization
Engineering support for ISA & algorithm questions	Manufacturing cost estimates for 7nm (informed by FPGA results)
Full NRE funding for Phase 1 ($67.5K budgeted)	Engineering time (optional — see cost models below)

Cost Sharing Models

Three Options — Dragon Ball Chooses

1. **Pure OEM Model:** Quillon funds everything. Dragon Ball contributes engineering time only. Zero capital risk for Dragon Ball. Dragon Ball earns manufacturing margins on 7nm production.
2. **Co-Development Model:** Dragon Ball provides FPGA boards and synthesis tools from existing inventory. Quillon provides RTL and firmware. Costs shared roughly 70/30 (Quillon/DB). Dragon Ball gets preferential manufacturing terms for 7nm.
3. **Joint Venture Model:** Equal investment, equal revenue share. Dragon Ball co-owns the QUG-V1 design. Highest commitment, highest upside.

We recommend starting with **Option 1 or 2** for the FPGA phase, then upgrading to Option 3 for 7nm if the partnership proves productive.

What We Need from Dragon Ball to Start

1. **FPGA device recommendation:** Which Xilinx/AMD part do you recommend for validating the full SoC (Xlattice + Xcrypto + RISC-V core)? We are open to Kintex-7, Kintex UltraScale, or Virtex UltraScale+.
2. **Existing infrastructure:** Do you have FPGA development boards from your ALPH/NEXA prototyping that we could target? Or should we design a new board?
3. **Reference data:** Can you share your ALPH FPGA prototype specs? (Resource utilization, clock speed achieved, power consumption.) This helps us calibrate our estimates against real-world numbers.
4. **Timeline estimate:** If we provide RTL by end of April, how quickly can your team synthesize, place-and-route, and test on an FPGA? (We expect 2–4 weeks for initial synthesis, 4–8 weeks for validation.)

Next Steps

NDA signed (if desired)	→	RTL architecture doc shared	→	FPGA device selected together	→	RTL delivered synthesis begins
Week 1		Week 2		Week 3		Week 4

If Chong is interested, we can share the RTL architecture document (block diagram, ISA encoding, memory map, bus protocol) this week. This is non-confidential technical specification — not trade secrets. We can sign a mutual NDA first if Dragon Ball prefers.

Technical Reference: QUG-V1 Architecture Summary

For convenience, here is the QUG-V1 SoC architecture relevant to FPGA prototyping:

Component	Description
CPU cores	16× RISC-V RV32IMC (8 mining + 2 PQ-crypto + 4 network + 2 management)
Xcrypto extension	Opcode space custom-0 (0x0B). Pipelined BLAKE3 — 9 cycles/hash at 1.5 GHz
Xlattice extension	Opcode space custom-1 (0x2B). 256-bit parallel modular arithmetic: `poly.mul`, `poly.reduce`, `ntt.fwd/inv`, `poly.add`
Memory	2 MB shared SRAM + DDR4 controller (for full-node blockchain storage)
Interconnect	Crossbar NoC, 128-bit data bus, split transaction
I/O	PCIe Gen3 x4, GbE MAC, USB-C (management + firmware update)
Target process	TSMC N7 (Phase 2) → TSMC N5 (Phase 3)
Power	9.5 W (N5 target), 12 W (N7 target)

Full technical details available in:

QUG-V1 Pocket Supercomputer Whitepaper (March 2026)
Genus-2 Jacobian VDF Mining Whitepaper (March 2026)
Dragon Ball × Quillon Response v2 (April 2026)

Why Dragon Ball Specifically

This is not a generic manufacturing RFP. Dragon Ball is uniquely positioned for this partnership:

- **Proven TSMC 5nm/7nm access:** Dragon Ball already manufactures KAS and ALPH miners on advanced nodes. No other potential partner has this relationship already established.
- **Thermal engineering solved:** Dragon Ball designs ASICs at 3,000W. The QUG-V1's 9.5W is trivial by comparison — zero thermal risk.
- **Mining distribution channels:** Dragon Ball already sells to the exact customer base that will buy QUG-V1 hardware. Immediate go-to-market.
- **FPGA prototyping infrastructure:** Dragon Ball's existing 355T boards and EDA toolchains can be reused, saving months.

No generic ODM (Flex, Jabil, Foxconn) can match this combination of silicon access, mining domain expertise, and distribution.

The Dual-Proof Advantage

Since the original proposal, Quillon has developed a **Dual-Proof mining system** that fundamentally changes the hardware landscape. The QUG-V1 is now the *only* hardware that can mine both proof types efficiently.

Hardware	BLAKE3 Path	Genus-2 VDF Path	Full Node
GPU (RTX 4090)	Efficient	Impossible (sequential)	No
High-end CPU (i9-14900K)	Moderate	Moderate	Yes (software)
Raspberry Pi 5	Slow	Very slow	Yes
Traditional ASIC	Dominant	Cannot adapt	No
QUG-V1 (Quillon)	**163× GPU**	**∼60× CPU efficiency**	**Yes (hardware)**

GPUs cannot mine the Genus-2 VDF path (inherently sequential). Traditional ASICs cannot adapt to algorithm changes (fixed-function silicon). CPUs can mine both paths but inefficiently. Only the QUG-V1 — with *both* Xcrypto and Xlattice on the same die — dominates both mining paths while running a full node. This creates a **captive hardware market** with no substitutes.

Genus-2 VDF Performance Projections

Platform	Doublings/sec	Power	Efficiency (d/s/W)
x86 CPU (i9 @ 5 GHz)	∼200	15W	13.3
ARM CPU (Cortex-A76 @ 2.4 GHz)	∼80	3W	26.7
GPU (RTX 4090, 1 core)	∼150	450W	0.33
QUG-V1 Xlattice core (projected)	**∼500–800**	**0.6W/core**	**833–1,333**

Note: QUG-V1 projections are based on architectural simulation, not silicon measurement. FPGA validation in Q3 2026 will provide measured performance data.

Strategic Value for Dragon Ball

Beyond manufacturing margin ($15–21/unit), this partnership offers Dragon Ball strategic advantages:

1. **Customer acquisition funnel:** Every QUG-V1 buyer at $149 is a potential future Dragon Ball ASIC customer at $3K–15K. The low entry price creates a massive new customer base.
2. **RISC-V capability:** Dragon Ball currently designs fixed-function ASICs. This partnership builds programmable SoC competence — reusable across future products.
3. **Network effect upside:** Unlike ASICs that depreciate as difficulty increases, QUG-V1 cards maintain value because they run full nodes and can adapt to algorithm changes via firmware. Dragon Ball can optionally embed a 1% developer fee in firmware for ongoing revenue.
4. **Diversification:** If the traditional ASIC market contracts (as it does cyclically), Dragon Ball has a consumer hardware product line as a hedge.
5. **First-mover advantage:** No competitor has a RISC-V mining SoC. Dragon Ball would be the exclusive manufacturer of a new product category.

IP Framework

Asset	Ownership
QUG-V1 RTL design (Verilog/SV)	Quillon Foundation — perpetual ownership
Xcrypto / Xlattice ISA extensions	Quillon Foundation — perpetual ownership
Mining firmware & blockchain software	Quillon Foundation — open-source (MIT license)
GDSII mask works (7nm/5nm)	Jointly held by funding party (negotiable per cost model)
PCB / enclosure design (DOCK)	Dragon Ball — owns mechanical/PCB design
Manufacturing license	Dragon Ball receives perpetual, royalty-free license to manufacture QUG-V1 **exclusively for Quillon**
Exclusivity window	Dragon Ball is sole manufacturer for 24 months post-launch
Third-party manufacturing	Not permitted during exclusivity window without mutual consent

These terms are proposed starting points. Final IP terms will be negotiated in a definitive agreement after FPGA validation.

Timeline & Risk Mitigation

Phase	Target	Worst Case	Mitigation
FPGA prototype	Q3 2026	Q4 2026	Start Phase 1A immediately; Dragon Ball's existing boards
7nm tape-out	Q1 2027	Q2 2027	Use mature N7 process (high yield); budget $500K for respin
First silicon validation	Q2 2027	Q3 2027	Start firmware dev on FPGA in Q3 2026, not after silicon
Volume production	Q3 2027	Q1 2028	7nm ships as revenue product even if 5nm is delayed
5nm upgrade	Q1 2028	Q3 2028	5nm is funded by 7nm revenue; not gating

If first 7nm silicon requires a respin, the timeline extends by ∼6 months and costs an additional $500K–1M. This risk is mitigated by: (1) thorough FPGA validation before tape-out, (2) using mature TSMC N7 with high yield, and (3) Dragon Ball's existing tape-out experience reducing integration errors.

Product Roadmap 2026–2030

Year	Product	Description
2026 H2	FPGA prototype	Validate Xcrypto + Xlattice on Dragon Ball's FPGA boards
2027 H1	QUG-V1 (7nm)	First silicon. 10 MH/s @ 12W. Credit-card form factor.
2027 H2	DOCK-8 / DOCK-16	Desktop enclosures. 8–16 card slots. $99–199 per dock.
2028 H1	QUG-V1 (5nm shrink)	17 MH/s @ 9.5W. Same package, drop-in upgrade.
2028 H2	QUG-V2	More cores, integrated Wi-Fi 7, lower power. DOCK-compatible.
2029+	QUG-V3	Next-gen architecture. Post-quantum integrated. 3nm target.

This is a **platform, not a one-off product**. Dragon Ball becomes the exclusive manufacturer for an entire product family spanning 5+ years.

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Quillon Foundation — April 2026

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